

RESPONSE TO THE REFEREE: NEURAL BELLMAN OPERATORS

QIAN QI
Peking University

Point-by-point response to the complete R21 Econometrica-level referee report. The revision incorporates the recovered R22 execution, new held-out R23 experiments, new proofs, exact provenance, and the original full-state accuracy objective.

1. RESPONSE TO THE COMPLETE R21 REFEREE REPORT

We thank the referee for distinguishing the now-reviewable local proof chain from the unresolved numerical requirements of the current-state Neural Bellman Operator. We respond to the final report on the complete R21 publication, not only to the earlier intermediate-snapshot report. The title, original stopped economy, admissible controls, author attribution, full state–time domain, and .01 objective are retained. The revision adds identification theorems, independently initialized continuum comparisons, an executed matched-initial-policy factorial, a held-out coordinate-control study, active-rejection tests, and a complete-domain witness refinement. We do not use regional success as evidence that the full-domain objective has been met.

The complete R22 scientific execution is recovered without changing its primary results. Its publication failures are documented separately. R23 adds 72 prospectively specified held-out configurations and two directed proof-instantiation audits. All old substantive text, mathematical arguments, numerical outcomes, and adverse comparisons remain in the manuscript’s dated historical appendix, supplement, and original repository paths. The primary narrative is reorganized rather than shortened by deletion.

1.1. *The principal methodological changes*

The old neural–Adam versus direct–L-BFGS-B comparison remains a valid practical baseline but is no longer used to identify representation. The primary crossed design fixes the delivered initial policy and runs both optimizers on both parameterizations. The new 47-coordinate direct control removes the budget’s redundant intercept without changing the financed family. A constructive interpolation proposition proves equality of the finite-node neural and direct policy classes. These changes identify an optimizer-conditional parameterization effect while leaving the stronger overall direct L-BFGS-B outcome visible.

The continuum proof no longer assumes identical initial slopes or preference drifts. A verified pair-moment bound controls the old mixture’s Jensen benefit. Independent corner seeds are used in both crossed executions. In the new held-out study the neural–Adam regional regret is below .003022, its uniform initial-to-final payoff gain exceeds .026360, and its uniform payoff advantage over both direct–Adam coordinate systems exceeds .003044. All of these are stopped-model continuum statements on the stated rectangle at time zero, not sampled-grid results or full-domain statements.

A separate exact-price theorem gives a unique financing root, an explicit oblique projection for its implicit derivative, a bracket including unbounded Gaussian tails, and a residual-to-root implication with the required exact-price error. All 72 delivered policies pass a directed bracket instantiation. A directed nonzero determinant certifies a concrete sixteen-node interpolation matrix. The ordinary proposal root residual and finite-difference tests are explicitly not treated as economic certificates.

2. SCIENTIFIC FINDINGS F1–F12

2.0.0.1. *F1: Regional accuracy is not full-state NBO accuracy.* We preserve this distinction in the abstract, introduction, current-state results, and conclusion. The regional network observes time, not current preference and wealth; its state dependence is supplied by the prescribed decoder and the four-subaccount construction. Separately, the retained current-state actor/witness refinement observes (t, u, x) and tightens the complete-domain, all-start-time certificate to 7.241462444 from 7.2783190635, on 1,024 fully checked MPFR cells. This is progress on the original object, not closure of .01 or a separately certified actor-payoff improvement. The main results state those limits next to the regional numbers.

2.0.0.2. *F2: The original representation ablation was confounded.* The referee is correct. The old comparison is now described as a matched delivered-policy-class end-to-end baseline. The new crossed experiments include neural–Adam, neural–L-BFGS-B, direct–Adam, and direct–L-BFGS-B, with matched initial policies and function/gradient accounting. R23 adds the direct-quotient coordinate control. The finite-node interpolation proposition rules out a difference in attainable raw arrays as an explanation. The comparison is about optimization geometry induced by parameterization, conditional on optimizer settings and budget; it is not a causal statement about all hidden-layer designs.

2.0.0.3. *F3: The direct configuration was stronger overall.* That fact remains in the main paper. Direct L-BFGS-B attains the sharper regional bound with fewer calls in all three new ensembles. The positive new result is narrower but identified: under the same Adam configuration and gradient budget, the neural systems have a strictly positive certified payoff advantage over both direct coordinate systems throughout the rectangle. This survives independent corner initialization and three held-out ensembles. We report it without asserting an unconditional neural accuracy–work advantage. The adverse neural L-BFGS-B ensemble above .01 is retained, not discarded.

2.0.0.4. *F4: The old continuum improvement used synchronized initialization.* The pair-moment Jensen theorem handles different initial slope and preference-drift vectors. Its proof controls weighted variances of consumption and preferences on the joint time–probability measure, includes correlation by Cauchy–Schwarz, and pays for the complement stopping event. Both new factorials use independent corner seeds. Each initial mixture receives its own Jensen upper bound; the strict improvement test subtracts that bound rather than assuming it vanishes. A simpler range-based theorem remains as a cross-check. This removes synchronized initialization as an assumption of the current continuum comparison.

2.0.0.5. *F5: Four-expert interpolation is not learned state generalization.* We agree and state the deployed subaccount policy explicitly. The new interpolation proposition is only a finite-time-node class-identification result; it is not a current-state approximation theorem. The cover-complexity proposition states sufficient regularity assumptions, domain and tolerance dependence, number of experts, and the cost of a verifiable cover. Runtime and storage are reported for the actual four-subaccount policy. Larger duplicated-expert timing tests are not called larger-domain experiments. Current-state learning is evaluated in its own full-domain refinement and is not inferred from these local mixtures.

2.0.0.6. *F6: The useful decoder is model-based.* The main text treats exact conditional pricing as an essential analytic feasibility layer, not as a learned hedge. It also clarifies that the

preference shock is partly unspanned; replicability of the selected factor-measurable consumption streams does not make the whole joint model complete. The verified residual-correction cone applies to a general controlled generator and supplies a constructive comparison beyond exact replication. Its theorem is retained with complete hypotheses and proof. The executed regional study still uses the chosen replicable family, and no incomplete-market sharp-accuracy experiment is claimed.

2.0.0.7. *F7: The unrestricted upper comparison is non-neural and essential.* The algorithm and cost ledger explicitly separate policy generation, fixed-policy evaluation, and unrestricted comparison. The inherited dual library’s 415.789-second construction cost is charged, whether unamortized or explicitly shared across solves. No new neural upper witness is credited with the regional sub-.0031 certificates. Actual pairwise policy-payoff ordering does not logically require that dual, whereas regret does. The proof-dependency map records this difference.

2.0.0.8. *F8: The global continuation witness remained inaccurate.* The new residual-correction construction and executed complete-domain refinement address the object directly rather than diagnosing only its boundary trace. The discounted suffix recursion controls every starting time, and the MPFR cover has no skipped cells. The resulting bound is smaller but remains well above .01. We retain that numerical outcome, the original target, and the older trace failures in the main comparison and historical appendix. We do not label this requirement closed or infer a global success from the local dual bypass.

2.0.0.9. *F9: Price transport is sensitivity, not broader state coverage.* The retained fixed-policy affine transport in k is now consistently labeled a sensitivity result on its original interval and initial-state region. It is not counted as current-state generalization or as an additional full-domain experiment. No price-node selection is used to improve the reported held-out regional outcomes.

2.0.0.10. *F10: The wealth increment changes the financing offset.* The current terminology is “sufficient re-budgeted wealth increment.” The exact counterfactual retains learned contrasts, slopes, and preference-drift shapes while changing the offset and checking the new stream. A new monotonicity proposition gives the economic ordering for the exact-price family when wealth nonexit holds. Its proof compares consumption and wealth pathwise at the common preference stopping time. The previous numerical .002 result retains its own verified intervals and small margin; it is not a minimum compensating variation or the welfare loss of an unchanged complete decision rule.

2.0.0.11. *F11: The gate had not exercised rejection and rollback.* The retained targeted stress design has four accepted and four rejected blocks. It verifies exact restoration of network state, optimizer state, and all three random-number states. Repeating a deterministic rejected proposal exposes stagnation. A separately identified supervisor alters the proposal schedule outside the restored state and produces six accepted and two rejected blocks. The failed primary financing proposal and the later root repair remain separately recorded. All 72 new R23 final proposals are accepted; consequently R23 is not presented as a replacement rejection-stress experiment or as a convergence theorem.

2.0.0.12. *F12: The replication study was not broad optimizer robustness.* The crossed optimizers, independently initialized corners, new held-out ensembles, and direct-coordinate control provide evidence that was missing from R21. Their scope remains stated: fixed architecture,

fixed Adam learning rate, fixed objective, and a 400-call cap. One new neural L-BFGS-B ensemble remains above .01. We make no architecture-wide, learning-rate-wide, dimension-wide, or unconditional superiority claim. The exact manifests and complete ledgers permit those scopes and failures to be audited.

3. TECHNICAL FINDINGS T1-T7

3.0.0.1. *T1: Provide one proof-dependency graph.* The supplement gives a table matching each headline assertion to its theorem and assumptions, exact script/function, result record, arithmetic backend, and proof-critical or diagnostic role. The publication manifest separately handles provenance, file hashes, and historical preservation; provenance is not substituted for proof validity.

3.0.0.2. *T2: MPFR is arithmetic redundancy, not mathematical independence.* This wording is retained. The independent arithmetic paths share the model derivation and several proof ingredients. Their consistency can detect arithmetic or implementation discrepancies, not independently establish the economic reduction. The new MPFR interpolation and continuous-price checks instantiate the stated proofs for frozen inputs.

3.0.0.3. *T3: Count objective, gradient, and line-search work.* Every configuration records function and gradient calls, accepted iterations, generation seconds, final-check seconds, root work, and parameter dimension. We explicitly label the inherited `line_search_evaluations` field as an extra-call proxy, $\max(0, N_f - 1 - N_{\text{iter}})$, not an instrumented count of rejected trials. No optimizer comparison equates Adam updates with L-BFGS-B iterations.

3.0.0.4. *T4: Match initial delivered policies.* The neural outputs at the sixteen nodes initialize the direct tensor exactly. The four primary arms are checked for binary64 equality of the delivered policy. The quotient representation has the same real-arithmetic policy; its floating-point discrepancy is recorded, and a nonidentical incumbent is independently certified. Thus an initialization mismatch is not silently absorbed into the representation effect.

3.0.0.5. *T5: Rename the old comparison.* The current narrative calls it a delivered-policy-class end-to-end baseline. Dated earlier exposition is retained verbatim and identified as historical. The new same-optimizer, matched-initial-policy comparisons and class-equivalence theorem carry the current representation interpretation.

3.0.0.6. *T6: Report online subaccount costs.* The supplement retains the quadrature-node, timing, and coefficient-storage ledger and explains that four subaccounts and conditional-price derivatives are deployed. The four-expert system uses 1,536 coefficient bytes and up to 8,192 quadrature nodes per query in the timing implementation. Larger duplicated systems test implementation scaling only. Online error certification and larger-state cover training are not inferred from latency measurements.

3.0.0.7. *T7: Distinguish preregistered execution and post-training certification.* The chronology records the exact R20, R22, and R23 protocols and source/result identities. R21 continuum/price/welfare audits and R23 continuous-price/rank instantiations are explicitly a posteriori. R22 repairs retain their post-observation status. The R23 seeds, budgets, coordinate arms, and continuum comparisons were fixed before their execution; all results are retained.

4. NEXT-ROUND REQUIREMENTS: ACHIEVED EVIDENCE AND REMAINING NUMERICAL TASKS

4.0.0.1. *N1: Crossed optimizer/representation experiment.* Executed twice with matched delivered initial policies, complete budgets and costs, plus a held-out redundant-coordinate control. The old confounded baseline remains historical rather than identifying.

4.0.0.2. *N2: State-conditioned execution.* A current- (t, u, x) actor/witness refinement is executed and fully covered. A policy-sensitive full-state payoff improvement and the .01 full-domain objective are not established by that refinement.

4.0.0.3. *N3: Independent initialization/general Jensen theorem.* The general pair-moment theorem and two independently initialized factorials remove the old synchronized-initialization requirement on the stated region.

4.0.0.4. *N4: Scaling beyond one rectangle.* A conditional cover-complexity theorem and measured implementation costs are supplied. No executed sharp high-dimensional or multicell original-economy frontier is claimed.

4.0.0.5. *N5: A neural regime that changes the frontier.* A reproducible, uniform, actual-payoff advantage is certified against both direct-Adam coordinates at matched gradient work. The best direct L-BFGS-B frontier remains stronger overall. We therefore distinguish this identified conditional effect from the stronger unresolved end-to-end requirement.

4.0.0.6. *N6: Full-state witness progress.* The complete-domain bound improves to 7.241462444 through an actual refinement and all-start-time check. This is quantitative progress, not achievement of .01.

4.0.0.7. *N7: Methodological identity.* The paper retains the Neural Bellman Operator and its original objective while identifying the learned generator, analytic feasibility layer, fixed-policy evaluator, and unrestricted comparison separately. No component is credited with another component's result. The new positive representation evidence is reported alongside the stronger classical baseline and outstanding global accuracy requirement, rather than changing the model or deleting the adverse evidence.

5. REVIEW OBJECT AND PRESERVATION

The canonical reading order is ECTA_R23.pdf, SUPP_R23.pdf, and RESPONSE_R23.pdf. The review entry and manifest identify the final immutable commit, source versions, all frozen result identities, exact checks, and PDF hashes. A new revision branch and a separately pinned referee-copy branch leave main and all older review/revision branches unchanged. The older navigation index is archived byte-for-byte before replacement. The compact main narrative is accompanied by complete dated historical appendices, not a reduced evidentiary record.